

NUnit Hands-On - UserManagerLib

1. Solution Structure

Solution: UserManagerLib.sln

Project 1 (given): UserManagerLib -> UserManager.cs

Project 2 (created): UserManagerLib.Tests -> UserManagerTests.cs

References added to UserManagerLib.Tests: UserManagerLib (project reference), NUnit, NUnit3TestAdapter (via NuGet Package Manager).

2. Source Project Code - UserManager.cs

```
using System;

namespace UserManagerLib
{
    public class UserManager
    {
        private string panCardNo;

        public string PANCARDNo
        {
            get { return panCardNo; }
            set
            {
                if (string.IsNullOrEmpty(value))
                {
                    throw new NullReferenceException("PAN Card number cannot be null or empty");
                }
                if (value.Length != 10)
                {
                    throw new FormatException("PAN Card number must be exactly 10 characters");
                }
                panCardNo = value;
            }
        }

        public string CreateUser(string panCardNo)
        {
            PANCARDNo = panCardNo;
            return "User created successfully";
        }
    }
}
```

3. Test Project Code - UserManagerTests.cs

```
using System; using
NUnit.Framework; using
UserManagerLib;

namespace UserManagerLib.Tests
{
    public class UserManagerTests
    {
        private UserManager userManager;

        [SetUp]
        public void Setup()
        {
            userManager = new UserManager();
        }

        [Test]
        public void CreateUser_ValidPANCard_ReturnsSuccessMessage()
        {
            var result = userManager.CreateUser("ABCDE1234F");

            Assert.That(result, Is.EqualTo("User created successfully"));
        }

        [Test]
        public void CreateUser_NullPANCard_ThrowsNullReferenceException()
        {
            Assert.That(() => userManager.CreateUser(null),
                Throws.TypeOf<NullReferenceException>());
        }

        [Test]
        public void CreateUser_EmptyPANCard_ThrowsNullReferenceException()
        {
            Assert.That(() => userManager.CreateUser(string.Empty),
                Throws.TypeOf<NullReferenceException>());
        }

        [Test]
        public void CreateUser_InvalidLengthPANCard_ThrowsFormatException()
        {
            Assert.That(() => userManager.CreateUser("ABC123"),
                Throws.TypeOf<FormatException>());
        }
    }
}
```

4. First Run - Test Result

Command used: dotnet test (or Test > Run All Tests in Visual Studio Test Explorer)

```
Test run for UserManagerLib.Tests.dll (.NETCoreApp,Version=v6.0)
NUnit Adapter 4.5.0.0: Test execution started
Running all tests in UserManagerLib.Tests.dll
```

```

NUnit3TestExecutor discovered 4 of 4 NUnit test cases

Passed!  CreateUser_ValidPANCard_ReturnsSuccessMessage
Passed!  CreateUser_NullPANCard_ThrowsNullReferenceException
Passed!  CreateUser_EmptyPANCard_ThrowsNullReferenceException
Passed!  CreateUser_InvalidLengthPANCard_ThrowsFormatException

Passed! - Failed:      0, Passed:      4, Skipped:      0, Total:      4,
Duration: 46 ms

```

5. Breaking the Test - Modified Source Code

The length check in PANCardNo setter is commented out in UserManager.cs to break the FormatException scenario:

```

set
{
    if (string.IsNullOrEmpty(value))
    {
        throw new NullReferenceException("PAN Card number cannot be null or
empty");
    }
    // if (value.Length != 10)
    // {
    //     throw new FormatException("PAN Card number must be exactly 10
characters");
    // }
    panCardNo = value;
}

```

6. Rerun - Test Result After Breaking Source

```

Running all tests in UserManagerLib.Tests.dll
NUnit3TestExecutor discovered 4 of 4 NUnit test cases

Passed!  CreateUser_ValidPANCard_ReturnsSuccessMessage
Passed!  CreateUser_NullPANCard_ThrowsNullReferenceException
Passed!  CreateUser_EmptyPANCard_ThrowsNullReferenceException
Failed!  CreateUser_InvalidLengthPANCard_ThrowsFormatException
Message:
    Expected: <System.FormatException>
    But was:  No exception thrown

Failed! - Failed:      1, Passed:      3, Skipped:      0, Total:      4,
Duration: 40 ms

```